University of Kufa

Faculty of Computer Science & Mathematics

Department: computer science

Course Name: (Advance Java)

Academic Year:2024-2025

Course Stage/Level: 1/ Second course

Date of Exam: 22/06/2025

Time: Three Hours

Final Exam

Note: Answer All questions only

Q1:- Answer Ten only from the following (10 Mark)

1- Consider the following code. What will be the output?

```
int[] arr = new int[5];
for (int i = 0; i < arr.length; i++) {
   arr[i] = i * i; }
System.out.println(arr[3]);
```

A) 6 B) 9 C) 3 D) 0

2- What is the result of this code?

```
int[] nums = {10, 20, 30};
System.out.println(nums[nums.length - 1]);
```

A) 10 B) 20 C) 30 D) Error

3- What does the following code output?

```
int[][] matrix = {
   {1, 2, 3},
   {4, 5, 6} };
System.out.println(matrix[1][2]);
```

A) 6 B) 5 C) 2 D) 3

4- Given this code, what is the output?

```
public class Test {
   static int add(int a, int b) {
      return a + b;   }
   static double add(double a, double b) {
      return a + b;   }
   public static void main(String[] args) {
      System.out.println(add(2, 3));
   } }
```

A) 5 B) 5.0 C) Compilation error D) 5.

5- What will be printed?

```
public class Example {
   void print(int a, int... b) {
      System.out.println(b.length);   }
   public static void main(String[] args) {
      new Example().print(1, 2, 3, 4);
   } }
```

A) 3 B) 4 C) 2 D) Error

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6- What is the result of the following?

```
String s1 = new String("Hello");
String s2 = "Hello";
System.out.println(s1 == s2);
```

A) true B) false C) Compilation error D) Hello

7- What is the output?

```
String str = "Java";
str.concat(" Programming");
System.out.println(str);
```

A) Java Programming B) Java C) Compilation error D) Programming Java

8- What is the output?

```
String str = "Java Programming";
System.out.println(str.substring(5, 8));
```

A) Prints Pro B) Prints ram C) Prints ogr D) Prints gra

9- Which of the following is NOT a valid string method in Java?

A) charAt() B) substring() C) equalsIgnoreCase() D) splitAt()

10- What will the following Java code output?

```
public class StringTest {
 public static void main(String[] args) {
 String s1 = "Java";
 String s2 = s1.concat("Script");
 String s3 = "JavaScript";
 System.out.println((s2 == s3) + " " + s2.equals(s3));
 }
}
```

A) true true B) false true C) true false D) false false

11- What is the output of the following?

A) 3 B) 4 C) 5 D) Compilation error

****

Q2: Select the correct code for five questions only (10 Marks)

1- Which of the following correctly declares and initializes a one-dimensional array in Java?

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A) int arr = new int[5]{1, 2, 3, 4, 5}; B) int arr[] = {1, 2, 3, 4, 5}; C) int arr[] = new int(5);
D) int[] arr = (1, 2, 3, 4, 5);

2- Which code correctly declares and accesses a 2D array element?

A- int[][] matrix = new int[3][3];
System.out.println(matrix(1)(2));

B- int matrix[3][3];
System.out.println(matrix[1][2]);

C- int[][] matrix = {{1,2},{3,4}};
System.out.println(matrix[1][1]);

D- int matrix = {{1,2},{3,4}};
System.out.println(matrix[1][1]);

3- Which of the following is a valid method definition in Java?

A- void method int a, int b) { return a + b; }

B- public void method(int a, int b) { System.out.println(a + b);}

C- function add(int x, int y) { return x + y; }

D- int add(x, y) { return x + y; }

4- Which code correctly gets the character at the 4th position of a String (index 3)?

A-String str = "Java";
System.out.println(str[3]);

B- String str = "Java";
System.out.println(str.charAt(3));

C- String str = "Java";
System.out.println(str.get(3));

D- String str = "Java";
System.out.println(str.at(3));

5- Which code correctly counts the number of vowels in a string?

A- String s = "Programming";
int count = 0;
for (char c : s.toCharArray()) {
if ("aeiou".contains(c))
count++; }

B- String s = "Programming";
int count = 0;
for (char c : s.toCharArray()) {
if("aeiouAEIOU".indexOf(c) != -1) count++;
} System.out.println(count);

C- String s = "Programming";
int count = 0;
for (int i = 0; i < s.length(); i++) {
char c = s.charAt(i);
if (Character.isVowel(c)) count++; }

D- String s = "Programming";
int count = s.chars().filter(c -> "aeiou".indexOf(c) > 0).count();

6- Which of the following correctly creates a string and compares it ignoring case?

A- String a = "hello";
String b = "Hello";
System.out.println(a == b);

B- String a = "hello";
String b = "Hello";
System.out.println(a.equalsIgnoreCase(b));

C- String a = "hello";
String b = "Hello";
System.out.println(a.equalsIgnoreCase(b));

D- String a = "hello";
String b = "Hello";
System.out.println(a.compareToIgnoreCase(b));0).count();

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Q3:- Answer five points (True , False) **(10 Marks)**

1- Java allows jagged arrays (arrays of rows with different lengths) using 2D arrays.
2- A multidimensional array in Java must have uniform row lengths.
3- Static methods can be called without creating an object of the class.
4- Two strings with the same content are always equal with = =.
5- You can use split() to divide a string using a regular expression.
6- str.trim() removes all white spaces from a string.

Q4: Select the right answer for five only **(10 Marks)**

1- Tests code comprehension and logical execution flow.

```
int[] nums = {10, 20, 30};

System.out.println(nums[1] + nums[2]);
```

A) 30 B) 40 C) 50 D) Compilation error

2- Match the method with its description:

A. equals()	1. Returns string length
B. length()	2. Compare reference identity
C. ==	3. Compare string content

3- Which of the following are valid method declarations in Java?
A) void test(int a) B) int test(int x, int y) C) test void(int a)
D) public void test(String s)

4- Arrange the following code lines to form a working method:

1. } 2-public static void greet() {
3- greet(); 4-System.out.println("Hello World");

5- Match each method call to its correct output:

Method Call	Output
str.length()	A. 'J'
"Java".charAt(0)	B. false
"abc".equals("ABC")	C. 4

رئيس القسم
أ. د فرقان حسن ربيع

مقرر المادة:
أ.م. د علي عبد الكريم حبيب

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Faculty of Computer Science & Mathematics

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Academic Year :2024-2025

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Date of Exam: 11/5/2025

Time: One Hour

Test B

حيدر علي عبد الزهرة

-A-

Q1:- Answer the following (5 Mark)

1- What is the output of the following code?

```
int[][] arr = {{1, 2, 3}, {4, 5, 6}}
System.out.println(arr[1][2])
```

A. 4
B. 5
C. 6
D. Error

2- Which statement is true about jagged arrays in Java?

A. All rows must have the same number of columns
B. They are illegal in Java
C. Rows can have different lengths
D. You can only create them with fixed size

3- Which of the following correctly uses a **professional for loop** to print all elements of a 2D array?

```
int[][] arr = {{1, 2}, {3, 4}};
```

A-
```
for (int i : arr)
    System.out.println(i);
```

B-
```
for (int[] row : arr)
    for (int val: row)
        System.out.println(val);
```

C-
```
for (int i = 0; i < arr.length; i++)
    for (int j : arr[i])
        System.out.println(j);
```

D- All of the above

4- What does the substring(2, 4) method return for the string "Computer"?

A) mp
B) mpu
C) put
D) mpu

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Test B

5- What is the output of the following code?

```
String str = "one,two,,four";
String[] parts = str.split(",");
System.out.println(parts.length);
```

A) 3
B) 4
C) 5
D) Compilation Error

****

Q2: Write java program to create array of 10 Names then passed this array to (5 Marks)

1- Find (Names [], Key) to find key("Ali") and print results in method.
2- Sort array using SortName(Names[]) then print sorted array
3- Change each first char of name in array to Uppercase( )
3- Use (**Switch**) to make options for call methods.

مقرر المادة:
أ.م. د علي عبد الكريم حبيب

رئيس القسم
أ. د فرقان حسن ربيع

University of Kufa
College of Computer & Mathematics Science
Department: Computer department
Subject Name: Advance Java

Final examination

Q1: Multiple choices (answer ten only):

1- Among the following which contains scanner class

a- Java.sql.timestamp b- java.io.time c-java.util

2- Find the output of the following program.

```
public class Solution{
    public static void main(String[] args){
        short x = 10;
        x = x * 5;
        System.out.print(x);
    }
}
```

a- 50 b- 20 c- compiler error

3- Select the valid statement.

a- char[ ] ch=new char(5) b- char[] ch=new char[5] c- char[ ] ch=new char[ ]

4- Select the valid statement to declare and initialize an array.

a- int [ ] A={ } b- int [ ] A={1,2,3} c- int [] [] A={1,2,3}

5- Find the value of A[1] after execution of the following program.

```
int[] A = {0,2,4,1,3};
for(int i = 0; i < A.length; i++){
A[i] = A[(A[i] + 3) % A.length];
}
```

a- 0 b- 1 c-3

6- equals () return result

a. Boolean b- int Value c- String

7- Identify the output of the following program.

```
String str = "abcde";
System.out.println(str.substring(1, 3));
```

a. abc b- bc c-cd

8- Identify the output of the following program.

```
Public class Test{
Public static void main(String args[]){
    String str1 = "one";    String str2 = "two";
  System.out.println(str1+str2);
} }
```

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Course Stage/Level: first
Lecturer Name: Ali A.A.Habib
Time: Two hours

Final examination

a- one b- onetwo c- twoone

9- Identify the return type of a method that does not return any value

a. int b. double c. void

10- Identify the output of the following program.

```
String str = " Hellow ";
System.out.println(str.Trim());
```

a- Hellow b- tHellow c- Hellowt

11- How many times will "Interviewbit" be printed.

```
Int count = 0;
do{
 System.out.println("Interviewbit");
 count++;
} while(count < 10);
```

a- 9 b-11 c-10

(10 Marks)

Q2:

Write java program to create three method , area of circle area,of square , area of triangle , with passing parameter from keyboard, use (switch) to make options for methods

(10 marks)

Q3: Describe two for the followings:

1- What is different between split( ) and substring( ) with example?
2- What is different between create string using String literal and new keyword with example?
3- What is different between method with return value and method with no return value ?

(10 Marks)

Q4: Write java program to create array Op[]={+, -, *, /, ^} then apply all these mathematic operations on variables (A, B).

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Good luck

رئيس القسم
أ.م. د. فرقان حسن صالح

مقرر المادة
أ.م. د علي عبد الكريم حبيب

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Test A

Q1:- Answer the following

1. Which of the following correctly creates and initializes a 2D integer array in Java?

A. int[][] matrix = new int[2][3];
B. int matrix = new int[2][3][];
C. int matrix[][] = new int[];
D. int[2][3] matrix = new int[][];

2. Identify the correct way to declare a jagged (irregular) 2D array:

A. int[][] jagged = new int[3][3];
B. int[][] jagged = {{1, 2}, {3, 4, 5}};
C. int[][] jagged = new int[3][3][3];
D. int[][] jagged = {{1, 2}, {3, 4}};

3. Which of the following is not a valid method for iterating over a 2D array in Java?

A. for (int i = 0; i < arr.length; i++) for (int j = 0; j < arr[i].length; j++)
B. for (int[] row : arr) for (int val : row)
C. while (i < arr.length) while (j < arr[i].length)
D. for (int i : arr) for (int j : arr[i])

4. What is the result of text.substring(2, 5) if text = "Elephant"?

0123456 7

A. epht
B. epha
C. epha
D. eph

5. What is the output of the following code?

0 1 2

String items = "pen,pencil,eraser";

String[] list = items.split(",");

System.out.println(list[2]);

A. pen
B. pencil
C. eraser
D. ,eraser

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Test A

****

****

(5 Marks)

Q2: Create a class called Invoice, this class have

1- **Create** array **items**[] ={ 1:sugar , 2:Tea, 3:rice }

2- Create array called **prices**

1	**200**
2	**400**
3	**600**

3- **Find Maximum and minimum price in <u>second column</u>**

4- ~~Create method~~ FindKey(items, prices, Key) to find **key="Tea"** , then print **its price** from prices array.

مقرر المادة:

أ.م. د علي عبد الكريم حبيب

رئيس القسم

أ. د فرقان حسن ربيع

اسم: أسعد سرور جاسم
الشعبة: (B)

First Test

Q1: Consider the following Java program. What will be the output when it is executed?

```
public class ReverseArrayLogic {
   public static void main(String[] args) {
      int[] arr = {3, 1, 4, 1, 5, 9};
      int sum = 0;
      for (int i = 0; i < arr.length / 2; i++) {
         int temp = arr[i];
         arr[i] = arr[arr.length - 1 - i];
         arr[arr.length - 1 - i] = temp;
      }
      for (int i = 0; i < arr.length; i++) {
         if (i % 2 == 0) {
            sum += arr[i] * 2;
         } else {   sum -= arr[i];   }      }
      System.out.println("Sum: " + sum);
   }
}
```

temp
arr[i
temp

{9, 5

output :- Sum: 9

}Q2: Write java program to create 2D array 3× 4 rows for students and columns for Grads of topics.

Find the failed student in the second topic.